Assessment Descriptor

Module Code: ITAG7105

Module Name: Data Mining

Semester: February to April 2026

Assignment: Individual Report (100%) on Descriptive, Predictive, and Text Analytics for Airline Customer Reviews

Submission Deadline: 11:59pm on 23rd April 2026

Maximum Word Count: 3,000 words (+/-10%) (excluding tables, figures, references, and appendices)

Assignment: Individual Report (100%)

Descriptive, Predictive, and Text Analytics for Airline Customer Reviews

Airlines operate in a highly competitive industry where customer satisfaction is critical for retaining loyalty and ensuring growth. Extensive historical data from passenger reviews has been collected to better understand customer satisfaction and improve airline services. You are tasked with proposing a model to predict whether a customer will recommend an airline or not, identifying the key factors influencing customer recommendations, and enhancing overall service quality.
The dataset includes:

> *Airline Name, Overall Rating, Review Title, The detailed text of the Review, Review Date, If the review was Verified, Type of Aircraft, Type of Traveller, Seat Type, Travel Route taken, Date Flown, Seat Comfort, Cabin Staff Service, Food & Beverages, Ground Service, Inflight Entertainment, Wifi & Connectivity, Value For Money, and Recommended.*

The target variable **'Recommended'** indicates whether the passenger has recommended the airline. The detailed report structure and technical tasks are presented below. The percentage marks for each section are shown in parentheses. All analytics tasks must be carried out using Python, and the results should be reproducible from the Python code, which must be submitted as an appendix. The indicated report structure must be followed precisely.

Report Structure and Technical Tasks:

1. **Introduction and Background (10%):**
   - You are required to introduce the business and analytical context of airline customer reviews, clearly articulating the importance of customer recommendation in the airline industry. The section should situate the problem within existing academic and practitioner literature on customer satisfaction, service quality, and predictive analytics.

The discussion should demonstrate awareness of contemporary challenges in analysing mixed-type data (numerical and textual) and justify the relevance of analytics in addressing these challenges.

2. Methodology (10%):

- Position your overall analytical approach within the CRISP-DM framework, clearly explaining how each phase informs the subsequent analysis.
- Justify the selection of descriptive statistics and visualisation techniques in relation to the characteristics of the cleaned dataset.
- Select two supervised machine learning algorithms suitable for predicting customer recommendation and provide:
  - A clear rationale for their selection,
  - Supporting academic and practitioner literature,
  - A critical discussion of their strengths and limitations in this context.

You are expected to demonstrate methodological reasoning rather than relying on generic or template-based explanations.

If Generative AI (GenAI) tools are used at any stage (e.g., for sentiment analysis, feature extraction, or code assistance), you must explicitly document:

- The purpose for which GenAI was used,
- How outputs were validated, adapted, or corrected using human judgement.

The focus of the methodology section should be on explaining the analytics tasks carried out, outlining your approach, and substantiating your narrative with pertinent literature. The technical tasks that must be completed are presented in Sections 3 and 4.

3. Data Pre-Processing (Data Cleaning and Preparation) (5%):

For **Numerical Data**:

- Identify and address non-trivial data quality issues, such as inconsistent ratings, logical conflicts between variables (e.g., Overall Rating versus Recommendation), or systematic missingness.
- Propose at least two alternative data cleaning or imputation strategies for one or more key variables. Implement one strategy and justify its selection over the alternative(s).
- Present a pre- and post-cleaning comparison for selected numerical variables and discuss how data preparation decisions may influence model outcomes.

For **Textual Data**:

- Design and justify a context-specific text preprocessing pipeline appropriate for airline review data.
- Critically reflect on limitations or biases introduced by preprocessing choices (e.g., loss of sentiment nuance, sarcasm, or negation).
- Present exploratory findings that are dataset-specific, such as recurring service issues or distinctive linguistic patterns, rather than generic word frequency outputs.

4. Data Analytics (40%):

4.a Descriptive and Predictive Analytics (Numerical features) (15%)

Descriptive Analytics:

- Compute descriptive statistics for all numerical features and identify at least two non-obvious or counterintuitive patterns.
- Develop a minimum of five visualisations, including:
  - At least two comparative visualisations,
  - At least one custom-designed visualisation intended for a non-technical airline management audience.
- Interpret all findings in the context of airline service quality and customer recommendation.

Predictive Analytics:

- Split the dataset into training and testing subsets and train the two selected supervised models.
- Tune one or two hyperparameters for each model.
- Conduct a model robustness or stress test, such as:
    - Training on one subset of airlines and testing on others, or
    - Training on earlier reviews and testing on later reviews.
- Evaluate models using appropriate performance metrics and critically discuss:
    - Generalisability,
    - Stability,
    - Practical suitability for decision-making.

4.b Descriptive and Predictive Analytics (Text features) (15%)

Descriptive Analytics:

- You may employ GenAI tools but you must:
    - Compare two distinct sentiment or emotion extraction approaches (e.g., LLM-based vs lexicon-based).
    - Manually review and critically analyse at least 10 ambiguous or misclassified reviews, explaining:
        - Why the model struggled,
        - What this reveals about airline review language.

Visualisation and Interpretation:

- Produce at least five visualisations, including those that explore contradictions (e.g., positive sentiment but low recommendation).
- Analyse and explain alignment or misalignment between text-derived features and numerical ratings.

Predictive Analytics:

- Train the previously selected supervised models using text-based features.
- Evaluate models in terms of:
    - Predictive performance,
    - Interpretability,
    - Managerial usefulness.

4.c Predictive Analytics Combining Numerical and Text Features (10%)

- Propose at least two feature integration strategies for combining numerical and textual data.
- Implement one strategy and justify its appropriateness.
- Assess the impact of feature integration on predictive performance.
- Identify which features dominate predictions and whether textual features add substantive value.
- Conduct a brief failure analysis where combined models underperform or behave unexpectedly.

5. Results and Discussion (15%):

- Present and compare model results using appropriate tables and metrics.
- Critically discuss limitations, assumptions, and analytical challenges.
- Reflect on:
    - The temporal relevance of findings,
    - Which insights are likely to remain valid if the data were collected today.

6. Conclusion and Recommendations (10%):

- Summarise key analytical findings and their implications for predicting customer recommendation.
- Provide data-driven managerial recommendations for airline decision-makers, clearly grounded in the analysis.

7. **Python Code (10%):**
   - Include the complete Python code used for all analyses as an appendix to the report. This should be presented directly embedded within the document. Avoid using non-editable formats like screenshots or images, as they hinder accessibility.

     Production-Grade Code Requirement (Mandatory if GenAI is Used)
     If GenAI tools are used for coding support, the submitted code must meet production-grade standards, including:
     - Modular structure (functions and/or classes where appropriate),
     - Clear, meaningful variable and function names,
     - Inline comments and docstrings explaining logic and assumptions,
     - Reproducibility (fixed random seeds, documented dependencies),
     - Error handling where relevant,
     - Logical separation of data loading, preprocessing, modelling, and evaluation steps.

     Please note that code that is copied, minimally edited, or lacks contextual adaptation will be penalised.

References: Please remember to include the list of references used at the end of your document, adhering to the Harvard referencing style. You are requested to use a minimum of 30 references in your assignment, and these should be a combination of academic and practitioner-based sources to ensure the quality and relevance of your research (wherein at least 10 references should be from the academic literature). Academic sources could include peer-reviewed journals, conference proceedings, and authoritative textbooks, while practitioner-based sources may encompass industry reports, reputable business publications, and case studies.

Additional instructions:

Please state on the front page:
- The module code and title
- The title of your assignment
- Your name and student number
- The actual word count

The maximum word count for assignment is 3,000 words (+/-10%) (excluding tables, figures, references, and appendices). Please note that a penalty will be applied for exceeding the word count by more than 10%.

Please ensure that the report follows the logical structure provided precisely, and that each section contributes to a comprehensive understanding of the analysis and its implications.

You are required to submit the assignment (as a Word file) via CANVAS by **11:59pm on 23rd April 2026**.

In addition, the individual assignment will be assessed using the postgraduate conceptual marking scale as recommended by the University (as outlined in appendix 1 and the Queen's Business School Postgraduate Student Handbook for further information).

The following criteria are also considered when assessing the assignment:
- Demonstrate wide reading and understanding of the assignment task.
- Ability to synthesise and critically evaluate relevant material.
- Quality and relevance of evidence/example presented to support position/claims.

- Structure including planning, organising, flow and coherence.
- References – quality of citations and correct style used.
- Overall presentation.

Plagiarism Detection Disclaimer:

Students agree that by taking this course, all required papers may be subject to submission for textual similarity review through the Canvas Plagiarism Framework (CPF) for the detection of plagiarism. All submitted papers will be included as source documents in the CPF reference database solely for the purpose of detecting plagiarism in such papers.

P.S.: This module is assessed 100% by an individual written report. Please also note that the dataset that has been given to be used for this assessment is based on an available dataset downloaded for educational purposes from a well-known source, Kaggle. The module teaching team does not own the dataset and hence, cannot comment on the accuracy of the data or be held liable for any misinformation that might be presented in the dataset. This is only to be used for educational purposes.
Note: Students must use the specific version of the dataset supplied by the module coordinator.

Use of AI in Assessments

In line with Queen's Business School (QBS) and Queen's University Belfast (QUB) guidance on Generative AI, students are permitted to use generative AI tools (e.g. Microsoft Copilot, ChatGPT) in an assistive role for this assessment.
As such, GenAI may be used to support the learning and writing process, for example to:

- help structure and organise the report,
- generate or refine draft text that is then substantially revised,
- support literature searching and summarisation,
- assist with coding, data handling, or debugging,
- proof-read and improve clarity and academic style.

However, all analysis, interpretation, judgement, ethical reasoning, and final conclusions must be the student's own work. Students are responsible for critically evaluating any AI-generated content and for verifying the accuracy of all facts, data, quotations, and references. Fabricated or incorrect sources generated by AI must not be used.
Students are required to include a written declaration after the title page stating whether and how Generative AI was used in the preparation of the assignment, in line with QBS guidance.

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Appendix 1: Conceptual Equivalents Scale Postgraduate

Module Descriptor	Mark Band	Criteria	Determinator within Grade Band
A (Outstanding)	80-100	• Thorough and systematic knowledge and understanding of module content • Clear grasp of issues involved, with evidence of innovative and original use of learning resources • Knowledge beyond module content • Clear evidence of independence of thought and originality • Methodological rigour • High critical judgement and confident grasp of complex issues	Originality of argument
A (Clear)	70-79	• Methodological rigour • Originality • Critical judgement • Use of additional learning resources.	Methodological rigour
B	60-69	• Very good knowledge and understanding of module content • Well-argued answer • Some evidence of originality and critical judgement • Sound methodology • Critical judgement and some grasp of complex issues	Extent of use of additional or non-core learning resources
C	50-59	• Good knowledge and understanding of the module content • Reasonably well argued • Largely descriptive or narrative in focus • Methodological application is not consistent or thorough	Understanding of the main issues
Marginal Fail	40-49	• Lacking methodological application • Adequately argued • Basic understanding and knowledge • Gaps or inaccuracies but not damaging	Relevance of knowledge displayed
Weak Fail	0-39	• Little relevant material and/or inaccurate answer or incomplete • Disorganised • Largely irrelevant material and misunderstanding • No evidence of methodology • Minimal or no relevant material	Weakness of argument